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United States Patent	12413309
Kind Code	B2
Date of Patent	September 09, 2025
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Autonomous mining system based on real-time digital video communication that uses ethernet and optical technologies

Abstract

An autonomous mining system includes a real-time digital video transmission sub-system configured to obtain video streams from underground, and transfer the video streams to a control center located above ground; and an exploration and maintenance sub-system located underground, and configured to extract a resource and bring the resource to the surface, based exclusively on commands received from the control center through the real-time digital video transmission sub-system.

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Appl. No.:	17/923960
Filed (or PCT Filed):	May 18, 2021
PCT No.:	PCT/IB2021/054282
PCT Pub. No.:	WO2021/234576
PCT Pub. Date:	November 25, 2021

Prior Publication Data

Document Identifier

US 20230188212 A1

Publication Date

Jun. 15, 2023

Related U.S. Application Data

us-provisional-application US 63027244 20200519

Publication Classification

Int. Cl.: **H04B10/116** (20130101); **E21C35/24** (20060101); **H04B10/67** (20130101); **H04N7/18** (20060101)

U.S. Cl.:

CPC **H04B10/116** (20130101); **H04B10/672** (20130101); **H04N7/181** (20130101); **E21C35/24** (20130101)

Field of Classification Search

CPC: H04B (10/116); H04B (10/672); H04N (7/181); E21C (35/24)

USPC: 398/172

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. National Stage Application of International Application No. PCT/IB/2021/054282, filed on May 18, 2021, which claims priority to U.S. Provisional Patent Application No. 63/027,244, filed on May 19, 2020, entitled “REAL-TIME DIGITAL VIDEO TRANSMISSION SYSTEM AND METHOD BASED ON INDUSTRIAL ETHERNET AND OPTICAL WIRELESS COMMUNICATION TECHNOLOGY FOR COAL MINE MONITORING, COMMAND AND CONTROL,” the disclosures of which are incorporated herein by reference in their entirety.

BACKGROUND

Technical Field

(1) Embodiments of the subject matter disclosed herein generally relate to a system and method for video communication in an environment in which there is no natural light, and more particularly, to using a combination of Ethernet and Optical protocols for video communication in such an environment for resource exploration.

Discussion of the Background

(2) Coal is the most abundant and widely distributed fuel resource in the world. In recent years, with the increase of mining depth and intensity, the number of coal-mine accidents in which human life is lost is increasing. Thus, there is a need to strengthen safety in production and improve the efficiency of early warning and rescue work in the mines. Of course, the same concerns are valid for any underground resource that require human presence in the mine.

(3) Real-time digital video monitoring and remote control based on various communication protocols is considered as an extremely promising technology. For example, there is the possibility of using real-time video monitoring and remote control through a combination of Ethernet and Wi-Fi technologies, which is a relatively fast and straightforward way to get information from coal mines. In addition, considering the characteristics of most coal mines (e.g., the ubiquity of solid-state lighting infrastructure, no interference of natural light, and the particularity of electromagnetic environment), light emitting diodes (LED) or laser diodes (LD)-based optical wireless communication (OWC) combined with Ethernet is expected to play a vital role in complex mine environments [1]. Compared with the conventional radio frequency communication technologies (e.g., leaky feeder communication and Wi-Fi wireless communication), the OWC technology has

unique advantages, such as high bandwidth, low latency, and electromagnetic interference immunity. In this regard, Chinese patent publication no. CN 105680942 [2] describes a visible light communication transceiver module for used in an underground mine.

(4) However, the availability of the video communication in real-time inside a mine would not be able by itself to solve the safety problems typically encountered in a mining environment. More than that is required for saving human lives.

(5) Thus, there is a need for a new system that is capable of exploring resources in a mine without human presence while being capable of being supervised, in real-time, by a human operator or an automated control center.

BRIEF SUMMARY OF THE INVENTION

(6) According to an embodiment, there is an autonomous mining system that includes a real-time digital video transmission sub-system configured to obtain video streams from underground, and transfer the video streams to a control center located above ground, and an exploration and maintenance sub-system located underground, and configured to extract a resource and bring the resource to the surface, based exclusively on commands received from the control center through the real-time digital video transmission sub-system.

(7) According to another embodiment, there is a method for autonomous mining exploration, and the method includes acquiring underground video streams with a real-time digital video transmission sub-system; transferring the underground video streams to a control center located above ground; analyzing the underground video streams for determining a position and a status of plural mobile nodes; generating commands at the control center with respect to future actions of the plural mobile nodes; transmitting the commands to the plural mobile nodes through a combination of Ethernet and visible light communication, VLC, protocols; and actuating the plural mobile nodes, exclusively based on the received commands, to execute the actions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present invention, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a schematic diagram of an autonomous mining system;

(3) FIG. 2 is a schematic diagram of an OWC transceiver module;

(4) FIG. 3 is a schematic diagram of an OWC-based video collection and transceiver module;

(5) FIG. 4 illustrates a diffused line-of-sight communication configuration;

(6) FIG. 5 illustrates a non-line-of-sight communication configuration; and

(7) FIG. 6 is a flow chart of a method for exploring a mine with plural mobile nodes that are operated in an autonomous mode.

DETAILED DESCRIPTION OF THE INVENTION

(8) The following description of the embodiments refers to the accompanying drawings. The same reference numbers in different drawings identify the same or similar elements. The following detailed description does not limit the invention. Instead, the scope of the invention is defined by the appended claims. The following embodiments are discussed, for simplicity, with regard to an autonomous mining exploration system that communicates in real-time with a control center and/or a surface operator using a combination of Ethernet and OWC communication protocols. However, the embodiments to be discussed next are not limited to a mining operation, but may be applied to other environments where there is minimal natural light interference.

(9) Reference throughout the specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with an embodiment is included in at least one embodiment of the subject matter disclosed. Thus, the appearance of the

phrases “in one embodiment” or “in an embodiment” in various places throughout the specification is not necessarily referring to the same embodiment. Further, the particular features, structures or characteristics may be combined in any suitable manner in one or more embodiments.

(10) According to an embodiment, a novel real-time digital video monitoring and remote control system, which is based on Ethernet and OWC, is configured to autonomously mine a given underground location, without the physical presence of an actual person. Thus, in case of mine accidents, no human life is in danger. To alleviate the link alignment issues between the various components of the system, diffused line-of-sight and non-line-of-sight communication configurations between OWC transceiver modules and OWC-based mobile video surveillance modules of the system is proposed. In one application, ultraviolet light-based non-line-of-sight communication devices are used. These features are now discussed in more detail.

(11) FIG. 1 illustrates the schematic diagram of an autonomous mining system **100** that includes a real-time digital video transmission sub-system **110**, which is based on Ethernet and OWC communication technologies, and an exploration and maintenance sub-system **130**, which is used underground to explore a resource, e.g., coal, in cooperation with the transmission sub-system **110**. Thus, the autonomous mining system **100** includes underground and ground parts.

(12) The transmission sub-system **110** has an above ground part **112** and a below ground part **120**. An above ground region **102** and an underground region **104** are schematically separated in FIG. 1 by line **106**. Thus, any element that is placed in the figure above line **106** is part of the above ground part **112**, and any element that is placed below the line **106** is part of the below ground part **120** or the exploration and maintenance sub-system **130**. The above ground part **112** includes a video surveillance system **114**, a remote control system **116**, and a communication network **118**, which may be an optical fiber. The video surveillance system **114** and the remote control system **116** may be located together in a control center **115**. The optical fiber **118** is used for uplink transmission of video streams and the downlink transmission of control commands for various mobile nodes.

(13) The video surveillance system **114** may include one or more computing devices (for example, a computer) having one or more displays that may be monitored by a human. The system **114** may also include one or more applications that automatically analyze received video streams and using artificial intelligence, determines a status of the components (e.g., the mobile nodes) of the exploration and maintenance sub-system **130**. The status of these components is then passed to the remote control system **116** so that a decision is made whether to instruct these components to change their status or let the components to continue to be in their original status. For example, if one component of the exploration and maintenance sub-system **130** is the mobile node **132** (e.g., drilling rig, forklift, bulldozer, grader, dump truck, or other known mining equipment), the remote control system **116**, with or without the intervention of the human operator, analyzes the status of the mobile node **132** received from the video surveillance system **114**, compares it with a pre-established operation plan, might verify the status of the neighboring mobile nodes to prevent a collision, and decides to either change or not the status of the mobile node. In one application, suppose that the mobile node is a dump truck and the video analysis determined that the dump truck is filled with ore. In this situation, the remote control system **116** instructs the dump truck **132**, through the below ground part **120**, to move from its current location to another location. At the same time, the remote control system **116** instructs a loader **132** to stop loading the dump truck, and either enter into an idle status, or move to load another dump truck. All these instructions are generated either based on a direct input from the operator of the system, or autonomously, based on pre-established rules that are stored in the remote control system **116**. For these reasons, the remote control system **116** includes computing units, memories, monitors, computer applications, etc.

(14) The video streams acquired by the underground part **120** of the video transmission sub-system **110** are moved to the above ground part **112** along an optical fiber ring network **122**, and the communication network **118**. The optical fiber ring network **122** collects all the video streams from

the various branches of the mine, along dedicated optical fibers **124**. For simplicity, the mine in FIG. **1** is shown having a single branch, but for practical mines, there are many branches at various levels underground. The optical fiber ring network **122** ensures that feeds from all the branches and all the levels are collected and sent to the surface equipment through the optical network **118**.

(15) FIG. **1** shows that the optical fiber **124** extends into a single branch of the mine and is connected to plural cameras **126** and plural visible light communication (VLC) modules **128**. The cameras **126** and the VLC modules **128** are part of the underground part **120**. The cameras **126** and the VLC modules **128** are physically attached to the walls or ceiling of the branch of the mine. The cameras **126** can be visible light cameras, infra-red cameras, etc. The VLC modules **128** can be implemented in various ways and a single module includes at least one OWC transceiver module **129**, which is optically connected to the optical fiber **124**. The OWC transceiver module **129** is used for data exchange with the mobile nodes **132**, where each of the mobile nodes includes a corresponding OWC-based video collection and transceiver module **134**.

(16) FIG. **2** is a schematic diagram of the OWC transceiver module **129** mounted on the top or side wall of the mine. The OWC transceiver module **129** includes a housing **202** that houses a light source **204**, a corresponding driver **206**, two amplifiers **208** and **210**, an analog-to-digital converter (A/D) **212**, a digital-to-analog converter (D/A) **214**, a microcontroller unit (MCU) **216**, a lens **220**, a photodetector **222**, a filter **224**, an Ethernet module **226**, and a multi-voltage output power supply **228**.

(17) FIG. **3** is the schematic diagram of the OWC-based video collection and transceiver module **134** installed on the mobile node **132**. The OWC-based video collection and transceiver module **134** includes a light source **302**, a corresponding driver **304**, three amplifiers **306**, **308**, and **310**, two A/Ds **312** and **314**, a D/A **316**, an MCU **318**, a lens **320**, a photodetector **322**, two filters **324** and **326**, a power source **328**, for example, a rechargeable battery, and a camera **330**.

(18) The light sources **204** and **302** may be white-light light emitting diodes, laser diodes or any other novel light sources for simultaneous illumination and OWC. The drivers **206** and **304** are used to make the corresponding light sources work in a linear range and superimpose the electrical signals on the light source. The amplifiers **208**, **210**, **306**, **308**, and **310** are used for signal amplification. The A/Ds **212**, **312** and **314** are used to convert analog signals into digital signals; the D/As **214** and **316** are used to convert digital signals into analog signals; the MCUs **216** and **318** are used for digital signal processing; the lenses **220** and **320** are used to adjust the size of the received light spot; the photodetectors **222** and **322** are used to convert the detected optical signals into electrical signals; the filters **224**, **324**, and **326** are used to remove the signal noise; the power sources **228** and **328** are used for voltage conversion and power supply. The Ethernet module **226** is used to transmit data between the optical fiber ring network **122** and the OWC transceiver module **129**. The camera **330** in FIG. **3** is used for video monitoring of the mobile node and its environment.

(19) The cameras **126** on the top or side walls of the mine monitor the general underground environment in real time and transmit corresponding first video streams **125** to the servers of the video surveillance system **114** in the control center **115** through the optical fiber ring network **122**. The real-time video will be displayed on the video monitoring screens. Then, the control center **115** can send control commands to the OWC transceiver modules **129** through the optical fiber ring network **122** according to the specific requirements. The OWC transceiver modules **129** located on the top or side walls of the mine send the control commands to the OWC-based video collection and transceiver module **134** installed on the mobile nodes **132**. The mobile nodes **132** perform the assigned tasks according to the control commands and transmit the collected video streams **127** to the OWC transceiver modules **129** through the OWC-based video collection and transceiver module **134**. The OWC transceiver modules **129** transform the received video streams **127** into Ethernet format and send them and some other feedback information to the servers of the video surveillance system **114** in the control center **115** through the optical fiber ring network **122**. Thus,

the control center **115** receives first video streams **125** collected by the cameras **126** and second video streams **127** collected by the VLC modules **128**. In addition, various other information may be collected by these elements, as the temperature, pressure, chemical air composition, etc. inside the mine.

(20) According to an embodiment, there is a communication protocol between the OWC transceiver module **129**, which is mounted on the top or side wall of the mine, and the OWC-based video collection and transceiver module **134**, which is installed on the mobile node **132**. In the downlink, the optical command signals **117** generated by the remote control system **116**, are transmitted successively to the Ethernet module **226** (as the Ethernet module is optically connected to the optical fiber **124**), then to the MCU **216**, D/A **214**, amplifier **208**, driver **206**, and finally to the light source **204**. Note that the Ethernet module **226** is configured to transform the optical command signals **117** into electrical signals **227** so that the processor MCU **216** receives electrical signals. The light source **204** emits optical signals **205** that are being transmitted through an underground air channel to the mobile nodes **132**. The optical signals **205** are focused into the photodetector **322** in the OWC-based video collection and transceiver module **134** by the lens **320**. The detected signals **205** are transformed into electrical signals **323** by the photodetector **322**, and are transmitted successively to the amplifier **308**, filter **324**, A/D **312**, and finally to the MCU **318**. The MCU **318** extracts the commands embedded into the electrical signals **323** and instructs the mobile node **132** to go to a designated place and perform tasks according to the control commands received from the remote control system **116**.

(21) In the uplink, the camera **330** of the OWC-based video collection and transceiver module **134** collects the local video at the designated location, for example, around the mobile node **132** and/or of the mobile node to determine a status of the node, and transmits the video stream to the amplifier **310**, filter **326**, A/D **314**, and finally to the MCU **318**. The MCU transmits the data to the D/A **316**, amplifier **306**, driver **304**, and finally to the light source **302** for emitting optical signals **303** that include the desired information regarding the mobile node **132** and its environment. The light source **302** sends the optical signals **303** (which includes the video stream **127**) through an air channel to the photodetector **222** of the OWC transceiver module **129**, which is mounted on the top or side wall of the mine. After transforming the received optical signals **303** into electrical signals **223** with the photodetector **222**, the electrical signals **223** are transmitted through the amplifier **210**, filter **224**, and A/D **212** to the MCU **216**. The signals may be processed at the MCU **216** prior to being sent to the ethernet module **226**, where the electrical signals are transformed into optical signals and transmitted along the optical fiber **124** and the optical fiber ring network **122** to the video surveillance system **114** in the control center **115**.

(22) The exploration and maintenance sub-system **130** includes, in addition to the plural mobile nodes **132**, one or more recharge stations RS, which are configured to either sit on the floor of the mine or be attached to the wall of the mine. The recharge stations are configured to recharge the power source **328** of the OWC-based video collection and transceiver module **134** and, if the case, the batteries that power the mobile nodes **132**. For example, the MCU **318** receives information from the power source **328** about its power level. When the power level is detected to be below a certain level, the MCU **318** instructs the mobile node **132** to move to the closest available recharge station, electrically connect to the recharge station, and recharge its power source to a minimum power level. These operations may be performed independent of the commands received from the control center **115**. The recharge stations RS are electrically connected to a power supply line **150**, as shown in FIG. 1. The power supply line **150** extends from a power generator **152**, which is located above ground and which is configured to supply electrical power. The power supply line **150** is also connected to various light appliances **154**, which are attached to the walls or ceiling of the mine. The power generator **152**, the power supply line **150**, and the various light appliances **154** are typically part of an existing mine, i.e., they are already there when the autonomous mining system **100** is deployed as each existing mine has such a power and light supply. Thus, the recharge

stations RS, the VLC modules **128**, and the cameras **126** can also be attached to the existing power supply line to supply them with electrical power.

(23) The autonomous mining system **100** uses the video streams **125** and **127** obtained from both the cameras **126**, which are fixedly attached to the walls or ceiling of the mine, and the cameras **330**, which are attached to the mobile nodes. In this way, the video streams **125** and **127** cover the entire field of operation of the mobile nodes, from various angles, which allows the artificial intelligence software applications of the control center **115** and/or the human operator that might be present in the control center to have an accurate and detailed picture of the mining operations underground. In one application, each mobile node has its unique identification ID, so that specific commands may be transmitted from the control center to the mobile node. The mobile nodes and the control center can use a combination of communication protocols, which includes a wired communication protocol and a wireless communication protocol. The processors MCU and the Ethernet module **226** are responsible for “translating” the transmitted data from one protocol to the other protocol, either way. In one application, the data communication between the control center **115** and the VLC modules **128** is achieved by using the Ethernet. Ethernet is defined herein as being a communication protocol to link and connect computing devices to form a Local Area Network or LAN. Every device with Ethernet connectivity will have a Network Interface Card (NIC) which will store the device's unique physical address or identifier. This physical address is also called MAC (Media Access Control) address. The Ethernet protocol comes in various speed variants like the Standard Ethernet (10 Mbps), Fast Ethernet (100 Mbps), Gigabit Ethernet (1000 Mbps) and 10 Gigabit Ethernet (10 Gbps). Other communication protocols may be used over the communication network **118**, optical fiber ring network **122**, and the optical fiber **124**.

(24) The wireless communication protocol used between the VLC modules **128** and the mobile nodes **132** may include any known one. In order to send data over such a wireless channel, a modulation of the light is required. A modulation is the form in which the light signal varies in order to represent different symbols in order for the data to be decoded. Unlike radio transmission, a VLC modulation requires the light signal to be modulated around a positive dc value, responsible for the lighting aspect of the lamp. The modulation will thus be an alternating signal around the positive dc level, with a high-enough frequency to be imperceptible to the human eye or even with a low frequency, as no humans are expected to be present inside the mine. Due to this superposition of signals, implementation of a VLC transmitter usually require a high-efficiency, higher-power, slower response DC converter responsible for the LED bias that will provide lighting, alongside a lower-efficiency, lower-power, but higher response velocity amplifier in order to synthesize the required ac current modulation. There are several modulation techniques possible for this kind of communication, forming three main groups: Single-Carrier Modulated Transmission (SCMT), Multi-Carrier Modulated Transmission (MCMT), and Pulse-Based Transmission (PBT). In one application, the VLC module **128** can be used to replace the light appliance **154**.

(25) Still with regard to the visible light communication channel, FIG. 4 shows the diffused line-of-sight configuration **410** and FIG. 5 shows the non-line-of-sight communication configuration **510** between the OWC transceiver module **129** and the OWC-based video collection and transceiver module **134**. Compared with the conventional point-to-point communication configuration, both of the configurations shown in FIGS. 4 and 5 can significantly alleviate link alignment issues. For a diffused line-of-sight communication system, the light source has a large divergence angle, which enables a large communication coverage. For a non-line-of-sight communication system, the light beams will arrive at the detector after being scattered by impurity particles in the air of the mine or reflected by the wall, ground, or other obstacles found in the mine. It is worth noting that the scattering effect will be enhanced in the coal mine due to the Mie scattering of the impurity particles. Moreover, as the ultraviolet light incurs lower path loss in non-line-of-sight OWC when compared to the visible light, the utilization of ultraviolet light in the coal mine for the non-line-of-sight communication is promising.

(26) The fact that the control center **115** receives general video streams **125** from the cameras **126**, which show large portions of the mine and the mobile nodes scattered in those portions of the mine, and also receives local video streams **127** from the cameras **330** located on the mobile nodes **132**, which show more detailed portions of the mine, right around the mobile node, allows the control center to receive a full and accurate picture of the mine. To enable the control center or the human supervising the control center to correctly instruct a mobile node to perform a certain action or to move to a certain location in the mine, in one embodiment, various marking elements **170** are distributed in the mine, either on the walls or on the floor **172** or on the ceiling of the mine, so that a spatial orientation in the mine can be established based solely on these marking elements. The marking elements may be placed as close as necessary to each other, for example, every 10 m, to also provide a depth impression for the operator of the control center or for the software applications that determine the next move of the mobile nodes. Further, the marking elements **170** may be implemented as simple mechanical objects having a known length and orientation, or they may have electronics **174** that are capable of emitting light, or other electromagnetic signals that can be captured by the VLC modules **128**, and/or cameras **126**, and/or cameras **330**, so that these elements can also get a depth impression about their environment.

(27) A method for autonomous mining exploration is now discussed with regard to FIG. **6**. The method includes a step **600** of acquiring underground video streams **125**, **127** with a real-time digital video transmission sub-system **110**, a step **602** of transferring the underground video streams **125**, **127** to a control center **115** located above ground, a step **604** of analyzing the underground video streams **125**, **127** for determining a position and a status of plural mobile nodes **132**, a step **606** of generating commands at the control center **115**, a step **608** of transmitting the commands to the plural mobile nodes **132** through a combination of Ethernet and visible light communication, VLC, protocols, and a step **610** of actuating the plural mobile nodes **132** exclusively based on the received commands.

(28) The plural mobile nodes are configured to extract a resource from underground and bring the resource to the surface. The resource may be any known resource, e.g., coal, copper, etc. The underground video streams include first video streams collected by cameras attached to a wall or ceiling of the mine, and second video streams collected by cameras located on the plural mobile nodes. In one application, the second video streams are transmitted from the plural mobile nodes to VLC modules exclusively through the wireless VLC protocol. The VLC modules and the cameras exchange data with the control center exclusively through the Ethernet protocol, thorough an optical fiber.

(29) The disclosed embodiments provide an autonomous mining system that is capable of coordinating the movement and functions of plural mobile nodes inside of a mine, with no human presence in the mine, so that the coal or any other resource present in the mine is autonomously extracted and brought to the surface. It should be understood that this description is not intended to limit the invention. On the contrary, the embodiments are intended to cover alternatives, modifications and equivalents, which are included in the spirit and scope of the invention as defined by the appended claims. Further, in the detailed description of the embodiments, numerous specific details are set forth in order to provide a comprehensive understanding of the claimed invention. However, one skilled in the art would understand that various embodiments may be practiced without such specific details.

(30) Although the features and elements of the present embodiments are described in the embodiments in particular combinations, each feature or element can be used alone without the other features and elements of the embodiments or in various combinations with or without other features and elements disclosed herein.

(31) This written description uses examples of the subject matter disclosed to enable any person skilled in the art to practice the same, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the subject matter is defined by the

claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims.

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(32) The entire content of all the publications listed herein is incorporated by reference in this patent application. [1] Naidu, K. P. S. S. V., Visalakshi, P. and Chowdary, P. C., 2019. Coal mine safety system using Li-Fi technology. [2] Chinese Patent Publication No. CN 105680942.

Claims

1. An autonomous mining system comprising: a real-time digital video transmission sub-system configured to obtain video streams from underground, and transfer the video streams to a control center located above ground; and an exploration and maintenance sub-system located underground, and configured to extract a resource and bring the resource to the surface, based exclusively on commands received from the control center through the real-time digital video transmission sub-system, wherein the real-time digital video transmission sub-system comprises: an optical wireless communication transceiver, attached to the underground, and configured to (1) receive commands from the control center at an Ethernet module, using an Ethernet protocol, and (2) optically communicate the commands with a light source, in a diffused light-of-sight or a non-line-of-sight configuration, to mobile nodes of the exploration and maintenance sub-system.
2. The system of claim 1, wherein the real-time digital video transmission sub-system comprises: an above ground part that is configured to analyze the video streams and issue the commands; and a below ground part that captures the video streams and also communicate using the optical wireless communication with the exploration and maintenance sub-system.
3. The system of claim 2, wherein the above ground part comprises: a video surveillance system that receives the video streams and provide status information about the exploration and maintenance sub-system; and a remote control system that, based on the status information received from the video surveillance system, instructs the mobile nodes of the exploration and maintenance sub-system to perform a given function.
4. The system of claim 3, wherein the given function includes one or more of moving from one point to another point, loading a truck, or drilling a hole.
5. The system of claim 2, wherein the below ground part comprises: an optical fiber ring network configured to receive optical signals from the above ground part; an optical fiber connected to the optical fiber ring network and configured to pass the optical signals; plural cameras configured to generate the video streams and to send the video streams to the optical fiber; and plural visible light communication, VLC, modules configured to optically communicate with the optical fiber and to wirelessly send the commands to the mobile nodes of the exploration and maintenance sub-system.
6. The system of claim 5, wherein the VLC modules include the optical wireless communication transceiver, the VLC modules communicate using the Ethernet protocol with the above ground part, and the optical wireless communication transceiver is configured to communicate in a wireless manner through ultraviolet light with the mobile nodes.
7. The system of claim 6, wherein the VLC modules transmit the commands from the above ground part directly to the mobile nodes.
8. The system of claim 5, wherein each VLC module comprises: the light source for sending the commands to the mobile nodes; a photodetector for receiving data from the mobile nodes; a processor for processing the commands and the received data; and the Ethernet module configured to transform electrical signals from the photodetector into a video stream to be injected into the optical fiber.
9. The system of claim 1, wherein the mobile nodes are configured to move and function underground without direct human intervention.
10. The system of claim 9, wherein the mobile nodes include one or more of a truck, drill rig,

loader, and grader.

11. The system of claim 9, wherein each mobile node of the mobile nodes includes: a photodetector for detecting the optical signal from a visible light communication, VLC, module associated with the real-time digital video transmission sub-system; a camera for acquiring a video stream to be sent to the real-time digital video transmission sub-system; a power supply; a light source for sending optical signals indicative of the video stream to the VLC module; and a processor that processes the optical signals and control the light source to send the video stream through a wireless visible light communication channel.

12. The system of claim 9, further comprising: recharging stations distributed throughout the mine and configured to recharge a power supply of the mobile nodes.

13. The system of claim 1, further comprising: a power supply located at the surface; and a power supply line that extends from the power supply to inside of the mine to supply electrical power.

14. The system of claim 13, wherein visible light communication modules and cameras distributed in the mine are electrically connected to the power supply line.

15. The system of claim 14, wherein the visible light communication modules and the cameras are also optically connected to an optical fiber that supplies the video streams to the control center.

16. A method for autonomous mining exploration, the method comprising: acquiring underground video streams with a real-time digital video transmission sub-system; transferring the underground video streams to a control center located above ground; analyzing the underground video streams for determining a position and a status of plural mobile nodes; generating commands at the control center with respect to future actions of the plural mobile nodes; transmitting the commands to the plural mobile nodes through a combination of (1) Ethernet protocol, between the control center and an optical wireless communication transceiver, which is attached to the wall or ceiling of the mine, and (2) optical light communication protocol, between the optical wireless communication transceiver and the plural mobile nodes; and actuating the plural mobile nodes, exclusively based on the received commands, to execute the actions.

17. The method of claim 16, wherein the plural mobile nodes are configured to extract a resource from underground and bring the resource to the surface.

18. The method of claim 16, wherein the underground video streams include first video streams collected by plural cameras attached to a wall or ceiling of the mine, and second video streams collected by cameras located on the plural mobile nodes.

19. The method of claim 18, wherein the second video streams are transmitted from the plural mobile nodes to VLC modules exclusively through the optical wireless protocol.

20. The method of claim 19, wherein the VLC modules and the plural cameras exchange data with the control center exclusively through the Ethernet protocol, through an optical fiber.
